Design Basis Accident — Military Action

LNG Terminal Attack

DBA-GAS-UC-LNG — Use Case

Issued Under: DBA-GAS Sector Framework

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Preface

This document is an L3 Use Case analysis issued under the Design Basis Accident — Military Action (DBA-MA) framework. It applies the DBA-MA methodology, established in the parent framework and adapted to the Oil and Natural Gas sector in the L2 child document, to a specific and consequential attack vector: deliberate military action against liquefied natural gas (LNG) terminal infrastructure.

The strategic context for this use case was established by events that have already occurred. The destruction of the Nord Stream I and II pipelines in September 2022 severed Europe's primary pipeline connection to Russian natural gas supply, accelerating a structural shift from pipeline dependency to LNG import dependency that was already underway. That shift concentrated strategic risk onto a new set of chokepoints: the LNG export terminals of the United States, Canada, and Qatar on one end; and the LNG import terminals, floating storage and regasification units (FSRUs), and connecting infrastructure of Europe on the other. What was once a supplemental supply pathway is now, for much of Europe, the primary one.

This use case addresses the threat to that supply chain from deliberate military action. It is written for a dual audience: U.S. critical infrastructure practitioners responsible for terminal security and resilience, and NATO defense and energy security planners responsible for allied supply assurance. The two audiences share a common interest in the outcome — reliable LNG supply as a component of allied energy security — but approach it from different institutional frameworks. This document attempts to bridge both.

The physical consequence envelope of an LNG terminal attack — pool fire, vapor cloud explosion (VCE), and boiling liquid expanding vapor explosion (BLEVE) — was held for separate treatment in the initial release of this document. That analysis is incorporated in Section 9 of this version. The strategic supply chain consequence remains the primary analytical thread; the physical consequence analysis provides the facility-level safety and emergency planning complement that USCG and emergency management audiences require.

1. Introduction

1.1 Purpose

This use case defines the Design Basis Accident scenario for deliberate military action against LNG terminal infrastructure. Its purpose is to bound the threat, trace the consequence chain with strategic supply disruption as the primary consequence domain, and identify the protective action thresholds and recovery architecture relevant to both U.S. and allied stakeholders.

As with all DBA-MA documents, this use case does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform. A terminal security program designed against a single-attack-vector threat will fail when a sophisticated adversary integrates maritime kinetic action, cyber-kinetic combined arms, and UAS modalities in a coordinated sequence. The DBA-MA approach defines the coordinated, multi-vector attack as the governing design event.

1.2 Scope and Applicability

This document addresses LNG terminal infrastructure in the following categories:

- U.S. LNG export terminals: large-scale liquefaction and export facilities on the Gulf Coast and Atlantic seaboard, including Sabine Pass, Freeport, Corpus Christi, Cameron, Cove Point, and Elba Island.
- Canadian LNG export terminals: facilities under development or in operation on the Pacific and Atlantic coasts, including LNG Canada.
- European LNG import terminals: regasification terminals and FSRUs providing the primary or supplemental natural gas supply to NATO member states, including Gate Terminal (Netherlands), Zeebrugge (Belgium), Świnoujście (Poland), and the network of FSRU deployments accelerated after 2022.
- Allied LNG supply chain: the tanker fleet, shipping lanes, and logistical infrastructure connecting export and import terminals.

This document addresses both the strategic supply chain and physical consequence envelopes of LNG terminal attacks. Physical consequence analysis — pool fire, VCE, and BLEVE scenarios — is developed in Section 9. LNG distribution infrastructure downstream of the import terminal (pipelines, storage, city gate stations) is addressed in DBA-GAS-FC1 (Natural Gas Transmission).

1.3 Relationship to Governing Documents

This document is a use case issued under the DBA-GAS Sector Framework, which is itself issued under the DBA-MA Parent Framework. The threat taxonomy, bounding methodology, and consequence analysis structure are inherited from those governing documents and applied here to the LNG terminal-specific context.

The 2015 Energy Sector-Specific Plan (DOE/DHS) provides the foundational overview of the sector. TSA Security Directives for maritime facility security, the Maritime Transportation Security Act (MTSA), U.S. Coast Guard LNG facility regulations (33 CFR Part 127), FERC jurisdiction over LNG import/export facilities, and PHMSA regulations governing onshore LNG storage and processing constitute the primary U.S. regulatory reference environment. NATO energy security planning documents and the European Commission's energy security frameworks are the relevant allied reference environments.

1.4 Document Positioning: U.S. and NATO Audiences

This document is written for simultaneous U.S. and NATO audiences. For U.S. practitioners, it provides a DBA-MA structured consequence analysis applicable to facility security planning, sector exercise design, and regulatory gap identification. For NATO planners, it frames LNG terminal vulnerability as an alliance-level energy security concern, with consequences that extend across member state supply commitments and military logistics.

The two audiences use different terminology for related concepts. Where terminology diverges, this document uses the U.S. critical infrastructure protection framework as the primary reference and notes NATO equivalents where relevant. The strategic supply chain consequence analysis in Section 5 is written primarily for the NATO audience, as it addresses the alliance-level implications that U.S. facility-level planning frameworks do not typically capture.

2. LNG Terminal Profile and Asset Taxonomy

2.1 Terminal Architecture

An LNG terminal is among the most capital-intensive and process-complex facilities in the energy sector. Export terminals perform liquefaction — cooling natural gas to approximately -162°C (-260°F) — thereby reducing its volume by a factor of approximately 600, enabling storage and transport. Import terminals perform regasification — warming LNG back to a gaseous state for injection into the transmission pipeline system. FSRUs perform regasification aboard permanently moored vessels, providing flexible import capacity without onshore liquefaction or storage infrastructure.

The critical asset components of an LNG terminal — those whose loss or degradation produces the most significant consequence — are:

Marine infrastructure. Jetties, berths, loading arms, and mooring systems that interface with LNG carriers. A typical large export terminal has two to four berths capable of accommodating vessels with capacities of 140,000 to 266,000 cubic meters. Damage to marine infrastructure prevents vessel loading or unloading, the proximate cause of supply chain disruption, even if onshore processing equipment is intact.

Liquefaction trains (export terminals). The refrigeration process units that cool natural gas to LNG. Large export terminals typically have multiple trains, providing some redundancy, but individual trains entail very high capital costs ($1-3 billion per train) and long replacement lead times. Train damage is the governing recovery constraint for attacks on export terminals.

Storage tanks—full-containment cryogenic tanks storing LNG between production and loading. Tank capacity typically ranges from 160,000 to 200,000 cubic meters per tank. Tanks are the highest-consequence targets from a physical safety perspective — a breach of tank integrity under the right conditions can create vapor clouds and BLEVE scenarios as in the companion use case. From a strategic supply perspective, tank damage reduces buffer inventory that sustains export operations during maintenance or disruption.

Control and safety systems. Distributed control systems (DCS) manage the cryogenic process, emergency shutdown systems (ESD), fire and gas detection systems, and tank gauging systems. These are the primary targets for cyber-kinetic attack. An adversary who achieves DCS access can manipulate process conditions — cryogenic temperatures, pressures, flow rates — in ways that cause equipment damage or create conditions for safety system failure without any physical presence at the facility.

Utilities and power supply. LNG terminal operations require reliable electrical power for compression, refrigeration, pumping, and safety systems. Large export terminals typically generate a significant portion of their own power using gas turbines fueled by the LNG process stream. Loss of utility power or turbine generation capacity disrupts operations independent of any damage to the primary process equipment.

2.2 Strategic Terminal Inventory

The following table summarizes the key LNG terminals relevant to this use case's U.S./NATO scope. It is not exhaustive — it identifies the terminals whose loss or degradation would have the most significant allied energy-security consequence.

Terminal	Country	Type	Strategic Significance
Sabine Pass	USA	Export	Largest U.S. LNG export facility; six trains; primary supplier to European and Asian markets
Freeport LNG	USA	Export	Second largest U.S. exporter; three trains; 2022 outage demonstrated European supply sensitivity
Corpus Christi	USA	Export	Three trains operational; expansion underway; significant European contract base
Cameron LNG	USA	Export	Three trains: Gulf Coast; European market contracts, including French and Japanese buyers
Cove Point	USA	Export	Atlantic Coast; sole East Coast export terminal; European supply pathway
LNG Canada	Canada	Export	First major Canadian LNG export facility; Pacific Coast; Asia-Pacific primary; NATO ally supply
Gate Terminal	Netherlands	Import	Primary NW European import hub; gateway for Germany, Belgium, and the Netherlands supply
Zeebrugge	Belgium	Import/Trans	Major European import and trans-shipment hub; connects to the UK and continental markets.

Terminal	Country	Type	Strategic Significance
Świnoujście	Poland	Import	Critical NATO eastern flank supply; primary Polish LNG import; Baltic region security
FSRU Network (DE/IT/NL/FI)	Multiple	Import (FSRU)	Post-2022 emergency deployment; Germany, Italy, Netherlands, Finland; rapid capacity but vulnerable

2.3 Regulatory Authority Crosswalk

U.S. Coast Guard (USCG). Primary maritime security authority for LNG facilities under MTSA. 33 CFR Part 127 governs waterfront LNG facilities. USCG approves facility security plans and conducts inspections. The USCG has both the jurisdiction and the enforcement authority that TSA lacks for pipeline physical security, making the maritime regulatory framework for LNG terminals materially stronger than the land-based pipeline framework.

FERC. Jurisdictional authority over LNG import and export terminal siting, construction, and operation under the Natural Gas Act. FERC approval is required for new export terminals and significant modifications to existing ones. Security is not a primary FERC mandate, though FERC engineering reviews examine the design of safety systems.

TSA. Surface transportation security authority; less directly relevant to LNG terminals than to land-based pipeline infrastructure, though TSA has a role in landside security coordination.

PHMSA. Safety jurisdiction over onshore LNG storage and processing under 49 CFR Part 193. Safety-focused; does not address the military-action threat tier.

Department of Energy. Export authorization authority — DOE authorizes LNG export volumes to free trade agreement (FTA) and non-FTA countries. In a supply-disruption scenario, DOE's export authorization framework is a tool for redirecting supply, though it was not designed as an emergency supply-management instrument.

NATO / European Commission. No direct security jurisdiction over individual terminals, but NATO energy security frameworks and EU energy security regulations establish the alliance-level planning context within which European terminal security sits. The EU's Gas Security of Supply Regulation (994/2010, revised) establishes supply standard obligations for member states that are relevant to consequence planning.

2.4 Strategic Significance: The Post-Nord Stream Environment

The destruction of Nord Stream I and II in September 2022 was not merely an infrastructure event — it was a strategic realignment. Europe's energy security architecture, built over decades on the assumption of a reliable pipeline supply from Russia, collapsed within a single winter season. The response was an accelerated pivot to LNG imports, the rapid deployment of FSRUs across multiple member states, and a structural increase in U.S. LNG export commitments to European buyers.

The consequence of that pivot is that LNG terminal infrastructure — U.S. export capacity in particular — now occupies a position in the allied energy security architecture that it did not occupy before 2022. When Freeport LNG sustained an accidental explosion and fire in June 2022, the resulting six-month outage of approximately 2 billion cubic feet per day of export capacity contributed directly to European gas price spikes at the worst possible moment in the energy crisis. That was an accident. The DBA-MA framework asks what a deliberate, coordinated attack on multiple export terminals — or on the import terminals that receive their product — would produce.

The answer, under the threat conditions defined in Section 3, is a supply disruption with alliance-level consequences: European heating security at risk during winter, NATO logistics fuel supply constrained, and member state governments managing the domestic political consequences of energy scarcity simultaneously with the military threat that motivated the attack. This is the strategic objective of a Tier 1 adversary. The attack on the terminal is the instrument; the degradation of alliance cohesion and military capacity is the objective.

3. Design Basis Threat: Military Action Against LNG Terminals

3.0 Demonstrated Threat: The Evidentiary Record

Three categories of demonstrated events establish the evidentiary foundation for the design basis threat defined in this section.

Nord Stream I and II. As documented in DBA-GAS-FC1 (Natural Gas Transmission), the September 2022 coordinated destruction of the Nord Stream pipelines demonstrated that strategic energy infrastructure can be attacked with precision explosive ordnance, with deliberate ambiguity in attribution, and with alliance-level consequences. The operational template — pre-positioning, coordinated execution, strategic consequence disproportionate to the physical damage — is directly applicable to LNG terminal attack planning.

Freeport LNG Outage (2022). While accidental rather than deliberate, the Freeport LNG fire and six-month outage demonstrated the sensitivity of European energy markets to disruptions at U.S. export terminals. A single terminal's loss had measurable effects on European gas prices. This is the accidental baseline; the DBA-MA design basis event is the deliberate, coordinated, multi-terminal version.

Houthi Maritime UAS Campaign. From late 2023 onward, Houthi forces in Yemen conducted a sustained campaign of maritime drone and missile attacks against commercial shipping in the Red Sea and Gulf of Aden. The campaign demonstrated that a non-state actor with access to relatively low-cost UAS and anti-ship missile technology could impose high costs on global shipping, rerouting vessels around the Cape of Good Hope and increasing transit times and insurance costs for LNG tankers. Applied to LNG terminal approach channels or vessels alongside terminals during cargo transfer, the same operational model represents a credible Tier 2 threat to terminal operations.

3.1 Threat Actor Taxonomy

Tier 1 — State Actor. Nation-states with advanced persistent threat capability, maritime special operations forces, and the ICS/cyber capability to execute combined arms attacks against LNG terminal infrastructure. Russia has demonstrated both the motivation and the operational willingness to attack European energy infrastructure, as evidenced by the Nord Stream attack. China has a broad pre-positioning capability in U.S. critical infrastructure networks. Iran has demonstrated offensive maritime capabilities in the Persian Gulf and cyber capabilities against energy-sector targets—all three present credible Tier 1 threats to LNG terminal infrastructure.

Tier 2 — Near-State Proxy. Paramilitary and hybrid warfare organizations operating under state direction. The Houthi maritime campaign is the most directly applicable Tier 2 model — a proxy force executing maritime attacks on energy supply infrastructure with state-provided weapons and intelligence support. Applied to LNG terminal approaches or LNG carrier shipping lanes, the Houthis' operational template is the Tier 2 LNG threat prototype.

Tier 3 — Sophisticated Non-State Actor. Addressed by existing maritime security and MTSA frameworks. Not the primary design basis for this document, though the physical and cyber protective measures developed for Tier 1 and 2 threats provide residual benefit against Tier 3 actors.

3.2 Attack Modality Taxonomy

3.2.1 Maritime Kinetic Action

Maritime kinetic attack is the foundational modality for LNG terminal threat analysis. It encompasses:

- Vessel-borne explosive attack on terminal marine infrastructure: jetties, loading arms, and moored LNG carriers during transfer operations. A carrier alongside a terminal with a partially transferred cargo load represents a uniquely high-consequence target — the vessel's own LNG inventory amplifies the consequence of a successful kinetic strike.
- Swimmer delivery vehicle (SDV) or combat diver attack on subsea infrastructure: jetty pilings, subsea pipelines connecting the terminal to the transmission grid, and mooring systems. Subsea attacks are difficult to detect, attribute, and respond to, consistent with the Nord Stream operational template.
- Standoff anti-ship missile or rocket attack from maritime or coastal positions: targeting liquefaction trains, storage tanks, or control buildings from a range that keeps the attacker outside the terminal security perimeter.

3.2.2 Cyber-Kinetic Combined Arms

Cyber-kinetic combined arms against LNG terminals follow the same template established in the L2 document for refinery attack, with terminal-specific characteristics:

- DCS compromise targeting cryogenic process control: manipulation of liquefaction train temperatures, pressures, or flow rates to cause equipment damage or create conditions for safety system actuation. The cryogenic process envelope is narrow — deviations from design parameters can cause brittle fracture of process equipment, compressor surge, or damage to heat exchangers, all of which can take months to repair.
- ESD and fire-and-gas system manipulation: inhibiting safety system response to allow a process excursion to develop before operators can intervene manually. The kinetic component of the attack may target the physical ESD panels or safety system controllers to complement the cyber manipulation.
- Terminal management system compromise: disruption of vessel scheduling, cargo management, and berth allocation systems to impose operational delays without physical damage. Less consequential individually, but relevant in combination with other modalities.
- SCADA compromise of subsea pipeline systems connecting the terminal to the transmission grid: pressure manipulation that causes pipeline damage or triggers protective shutdowns, isolating the terminal from its supply or delivery network.

3.2.3 UAS — Aerial, Maritime Surface, and Subsea

Unmanned aerial, surface, and subsea systems represent a named attack modality in the DBA-MA threat taxonomy for LNG terminals. UAS has moved from an emerging to a demonstrated threat vector for energy infrastructure, and the LNG terminal environment presents specific UAS vulnerability characteristics.

Aerial UAS. Commercially available or militarized aerial drones provide standoff reconnaissance capability against terminal layouts, identifying process equipment, control building locations, and security postures without triggering conventional perimeter security. Armed aerial UAS — demonstrated at scale in the Ukrainian conflict and in Houthi operations — can deliver precision munitions against exposed process equipment, control buildings, or LNG carriers at a jetty. The relatively open airspace over coastal terminal locations, combined with the difficulty of distinguishing armed from reconnaissance drones at a distance, creates a detection-and-response challenge that conventional physical security perimeters do not address.

Maritime Surface UAS. Uncrewed surface vehicles (USVs) carrying explosive payloads represent the direct maritime application of the Houthis' operational model. USVs can be deployed from a standoff vessel or coastal position, navigate autonomously or under remote control to a terminal jetty or a moored LNG carrier, and deliver a shaped charge or bulk explosive against the hull or jetty structure. Current maritime security detection systems —

radar, AIS, and visual surveillance — were designed for conventional vessel threats and have limitations against small, low-radar-cross-section USV targets approaching at surface level.

Subsea UAS. Uncrewed underwater vehicles (UUVs) provide the subsea equivalent of the Nord Stream attack capability. A UUV can be pre-positioned in terminal approach channels, navigate to jetty pilings or subsea pipeline interfaces, and deliver a charge against infrastructure that is inaccessible to conventional security patrols. UUV detection requires active sonar, underwater surveillance systems, or physical inspection — capabilities that are not uniformly deployed at LNG terminals in the current security posture.

The three UAS modalities are most consequential when integrated into the cyber-kinetic combined-arms attack. Aerial UAS provides pre-attack reconnaissance, improving targeting precision for maritime kinetic action. Surface UAS delivers a kinetic effect against marine infrastructure while the cyber component suppresses terminal safety systems. Subsea UAS attacks the infrastructure that would otherwise sustain terminal operations during recovery from the surface attack. The integrated UAS-cyber-kinetic sequence is the design basis modality for Section 4.

3.2.4 Logistics and Supply Chain Denial

Tanker fleet targeting. LNG transport depends on a specialized global fleet of approximately 700 LNG carriers. Sustained interdiction of tanker shipping lanes — through maritime UAS attack, anti-ship missile threat, or mine-laying — can disrupt supply chain delivery independent of any damage to terminal infrastructure. The Houthi Red Sea campaign demonstrated the effectiveness of this approach against commercial shipping; applied to LNG carrier routes between U.S. export terminals and European import terminals, it represents a supply chain denial strategy that does not require penetrating terminal security perimeters.

Equipment and spare parts interdiction. LNG liquefaction train components — compressors, heat exchangers, cryogenic pumps — have highly concentrated supply chains. Major compressor manufacturers are geographically concentrated. A Tier 1 adversary with supply chain access can pre-position against the parts needed for post-attack recovery, extending the effective disruption period beyond the physical damage timeline.

3.3 The Bounding Logic

The following parameters define the Design Basis Threat for LNG terminal attack:

- Adversary tier: Tier 1 state actor, with Tier 1 cyber pre-positioning and Tier 2 maritime execution providing plausible deniability.
- Modality: Integrated cyber-kinetic combined arms with UAS components — aerial reconnaissance preceding, maritime surface UAS delivering kinetic effect against marine infrastructure, subsea UAS targeting jetty or pipeline foundations, and DCS compromise suppressing safety system response.

- Target selection: Marine infrastructure (to prevent vessel operations) and liquefaction trains (to prevent production) at a minimum of two major U.S. export terminals simultaneously, coordinated with the disruption of European import terminal operations.
- Timing: Winter season, coordinated with elevated European gas demand, minimum storage inventory, and constrained alternative supply routing.
- Simultaneity: U.S. export terminal attack coordinated with European import terminal disruption, tanker route interdiction, and — potentially — concurrent action against the transmission pipeline system feeding the export terminal, as analyzed in the L2 document Scenario A.

3.4 Simultaneity and Compounding Effects

The simultaneity trap identified in the L2 document applies with particular force to the LNG terminal threat. The design basis event is not a single terminal attack — it is a coordinated sequence that exceeds the supply chain's adaptive capacity across multiple layers simultaneously.

An adversary that attacks a single U.S. export terminal assumes the global LNG market will reroute supply from other terminals to compensate. An adversary who simultaneously attacks multiple export terminals, interdicts tanker shipping lanes, and disrupts European import terminal operations has removed the adaptive mechanisms that would otherwise contain the consequences of any single event. The market cannot reroute supply when the supply nodes, the delivery nodes, and the transport pathway are all under simultaneous pressure.

The compounding effect extends to the land-side pipeline system. An attack on the natural gas transmission infrastructure feeding an export terminal — executed concurrently with the terminal attack — eliminates the terminal's supply input while degrading its export capacity. The terminal becomes unable to receive gas from the domestic transmission system and unable to load LNG onto waiting carriers simultaneously. This integrates the L2 Scenario A threat with the L3 LNG terminal threat into a single, coordinated campaign.

4. Design Basis Accident Scenario: LNG Terminal Attack

4.1 Initiating Event

Coordinated integrated attack against two major U.S. Gulf Coast LNG export terminals, executed during peak European winter demand. The attack sequence integrates three simultaneous components:

- Cyber component: Nation-state adversary, having achieved persistent DCS access through a multi-stage intrusion campaign over the preceding months, initiates coordinated manipulation of liquefaction train process parameters at both terminals simultaneously. The ESD system inhibits commands from being executed. Fire and gas detection alarm suppression is activated. Terminal control room operators observe anomalous data that is partially masked by the cyber operation.

- Maritime UAS component: Unmanned surface vehicles, pre-positioned in the approach channels outside the established security zones, navigate to the jetty structures at both terminals. Shaped charge detonation against the loading arm foundations and jetty berthing infrastructure at each terminal. Simultaneous aerial UAS strike against exposed electrical switchgear and control building communications infrastructure at one terminal, degrading the operator's ability to coordinate emergency response.
- Subsea component: Pre-positioned UUVs detonate against the subsea pipeline risers connecting each terminal to the offshore pipeline network, isolating the terminals from their feed gas supply independent of any onshore pipeline attack.

The three components are timed to initiate within a window of less than 15 minutes, exploiting the gap between the initial alarm and the effective coordinated response.

4.2 Progression Sequence

T+0: Simultaneous cyber initiation, USV detonations, aerial UAS strike, and UUV detonations. Control room operators at both terminals receive an alarm flood partially masked by cyber suppression. Emergency shutdown initiated — partial success, as ESD inhibits commands from the cyber component, preventing a full automated response at some subsystems.

T+0 to T+30 minutes: Marine infrastructure damage assessment underway. Loading arms at both terminals are damaged or destroyed — no LNG carrier can berth or complete transfer operations. Liquefaction train anomalies are developing due to DCS manipulation; operators are working to determine the extent of the control system compromise. Subsea pipeline isolation confirmed — feed gas supply to both terminals interrupted.

T+30 minutes to T+4 hours: Emergency response activated at both terminals. USCG maritime security zone established. TSA and CISA notified. LNG carriers diverted from the approach. Liquefaction trains shut down—either by automated safety systems where ESD was not inhibited or by manual operator action—the extent of DCS compromise is under active forensic investigation. Physical damage to marine infrastructure assessed — both terminals are unable to conduct vessel operations.

T+4 to T+24 hours: DOE ESF-12 activation. CISA coordinates federal cyber response. National Security Council notified. Allied governments were notified through diplomatic channels. European import terminal operators are alerted to the incoming supply disruption. LNG spot market responds to news of terminal attacks — significant price spike. LNG carriers at sea divert to alternative loading ports where available.

T+24 to T+72 hours: Supply disruption propagates to European markets. Alternative supply routing analysis underway — Qatar, Norway, Australian LNG. Shipping lane diversion assessed. FSRU operators in Europe place emergency procurement orders. Storage drawdown accelerated.

NATO energy security coordination activated. U.S. government assesses attribution and considers response options.

T+72 hours and beyond: Recovery timeline governed by: marine infrastructure repair (months for loading arm replacement); liquefaction train integrity assessment and restart (weeks to months depending on process equipment damage); DCS forensic investigation and system reconstitution (weeks); subsea pipeline repair (months). European markets are managing supply deficits through storage drawdown, demand reduction, and alternative supply procurement for the duration of the repair period.

4.3 Consequence Envelope: Strategic Supply Chain

Immediate supply disruption. Two major U.S. export terminals, representing approximately 25-30 percent of total U.S. LNG export capacity, are immediately offline. LNG supply to European buyers under long-term contracts from these terminals is interrupted. Spot market tightening begins within hours of attack confirmation.

European market consequence. European gas storage drawdown accelerates to compensate for reduced LNG import supply. Depending on storage inventory position and weather severity, storage may sustain markets for weeks to months. Price escalation in European gas markets affects electricity prices, industrial production costs, and household energy bills across multiple member states simultaneously.

NATO logistics consequence. NATO member-state militaries operating under an elevated readiness posture — the context in which an adversary would most likely execute this attack — have higher fuel requirements. Reduced natural gas availability affects the reliability of electricity generation in member states that depend on gas-fired generation, which in turn affects military base power supply, communications infrastructure, and logistics coordination systems.

Alliance cohesion consequence. The domestic political pressure created by energy scarcity during a security crisis is the adversary's strategic objective. Governments managing heating fuel shortages and electricity price spikes simultaneously with a military threat from the same adversary face internal political pressures that may constrain alliance solidarity and response options. This is not an accidental side effect of the attack — it is the design.

4.4 Protective Action Window

The cyber pre-positioning phase — typically lasting months — is the primary detection window. Network anomaly detection on ICS systems, anomalous vendor access to DCS historian data, and unusual process parameter read activity are the leading indicators available to terminal operators and to federal monitoring programs.

The maritime UAS threat presents a different detection window. USV and aerial UAS approaches can be detected by radar, electro-optical surveillance, and acoustic systems if those systems are deployed with sufficient coverage of the terminal approach zones. The current

USCG-required facility security plan framework addresses vessel traffic monitoring but was not designed for USV threats with the radar cross-section characteristics of current commercial and military systems.

UUV detection requires either active sonar or physical inspection programs that cover the approach channels and subsea pipeline routes. This is a significant capability gap at most commercial LNG terminals in the current security posture.

The integrated attack sequence described in Section 4.1 is designed to execute faster than the current detection-to-response cycle. Once the attack initiates, the protective action window is measured in minutes. The investment case for pre-attack detection and hardening is therefore orders of magnitude more favorable than that for post-attack response.

4.5 Recovery Timeline

- Marine infrastructure (loading arms, jetty structures): 6 to 18 months for major loading arm replacement, depending on equipment availability and fabrication lead times. Loading arms are custom-fabricated for each terminal configuration and are not shelf-available items.
- Liquefaction train assessment and restart (no major process damage): 4 to 12 weeks for DCS forensic investigation, system reconstitution, safety case re-validation, and regulatory clearance.
- Liquefaction train repair (process equipment damage): 6 to 24 months, depending on component damage. The replacement of the heat exchanger and compressor has extended lead times under normal supply chain conditions.
- Subsea pipeline repair: 3 to 12 months depending on water depth, sea state, and repair vessel availability. The same repair-vessel constraints that extended Nord Stream recovery planning apply to terminal subsea-pipeline repair.
- DCS and safety system reconstitution: 4 to 16 weeks for forensic investigation, system rebuild, and testing. This timeline governs the restart of process operations even if physical equipment is undamaged.

The governing recovery constraint is marine infrastructure — a terminal cannot export LNG without functional loading arms and jetty structures, regardless of the state of liquefaction, storage, or pipeline systems. Loading arm replacement is therefore the critical path item for export terminal recovery.

5. Strategic Supply Chain Consequence Analysis

5.1 U.S. Export Capacity Disruption

The United States has become the world's largest LNG exporter, with nameplate export capacity of approximately 14 billion cubic feet per day across operating terminals. This capacity is concentrated at a small number of facilities — six major export terminals account for the majority of total U.S. LNG export capability. The geographic concentration on the Gulf Coast, while reflecting the proximity of Permian and Appalachian gas supply, creates a threat environment in which a small number of simultaneous attacks can produce a disproportionate reduction in national export capacity.

The Freeport LNG accident in June 2022 demonstrated the European market's supply sensitivity to individual U.S. terminal disruptions. That event removed approximately 2 billion cubic feet per day of export capacity for approximately six months. Under the design-basis scenario — two major terminals simultaneously offline — the supply impact is approximately three to four times larger, occurring at a moment of deliberate adversary selection (peak winter demand, minimum storage, constrained alternative supply) rather than at the accidental summer-season timing.

DOE's export authorization framework was designed to manage commercial supply decisions, not emergency supply crisis response. The framework does not include pre-authorized mechanisms for the emergency redirection of LNG supply from Asian markets to European markets, or for the expedited authorization of additional supply from alternative sources during a supply emergency. This regulatory gap has direct consequences for the speed and effectiveness of the U.S. government's response to an attack on an export terminal.

5.2 European Import Dependency and Allied Supply Commitments

The post-2022 European energy security architecture rests on three supply pillars: Norwegian pipeline gas, LNG imports (primarily from the U.S. and Qatar), and residual pipeline supply from non-Russian sources. LNG imports have grown from a supplemental to a primary supply component for multiple member states, and the FSRU deployments of 2022-2023 created import capacity in countries — notably Germany — that had no LNG import infrastructure before the Ukraine conflict.

The FSRU fleet is the most vulnerable component of the European import architecture. FSRUs are permanently moored vessels performing regasification — they combine the vulnerability of a vessel at anchor with the critical infrastructure consequences of an onshore regasification terminal. They are typically moored in locations chosen for logistical convenience rather than security; their mooring systems are exposed to maritime UAS attacks, and their vessel classification means their security framework sits between maritime and onshore critical infrastructure frameworks, creating regulatory gaps.

EU Gas Security of Supply Regulation obligations require member states to maintain supply to protected customers — households, hospitals, and essential social services — under defined supply-stress scenarios. The design-basis attack scenario exceeds the supply-stress assumptions these regulations were designed to address. A member state that is simultaneously managing a military threat from the attacking adversary and a heating fuel shortage affecting its civilian population is in a position that the supply security regulations did not contemplate, and for which the supply diversification and infrastructure interconnection measures required by those regulations are insufficient.

5.3 NATO Logistics Implications

NATO military operations under Article 5 conditions depend on fuel supply chains that, in turn, depend on the energy infrastructure the alliance is defending. Natural gas powers electricity generation that powers base operations, communications, logistics coordination, and weapons system maintenance across member state military installations. Petroleum products refined from crude supply — itself dependent on pipeline and terminal infrastructure — fuel the tactical and strategic mobility that gives NATO's defense-in-depth posture its operational effectiveness.

An adversary that simultaneously degrades U.S. LNG export capacity, European gas supply, and the petroleum product pipeline infrastructure documented in the L2 child document has created a logistics environment in which NATO's military response capacity is constrained even as the strategic threat requiring that response has escalated. This is not a coincidental compounding of independent events — it is the logic of a campaign designed to degrade alliance response options before the primary military action.

NATO's energy security planning has historically focused on supply disruption as an economic and humanitarian concern, addressed through civilian emergency coordination mechanisms. The DBA-MA framework treats supply disruption as a military logistics concern, addressable through alliance defense planning processes. The gap between these framings — and the planning and investment that fall within them — is one of the most significant vulnerabilities in the current allied energy security architecture.

5.4 Spot Market Cascade Effects

The global LNG spot market is the primary adaptive mechanism that allows supply to flow around disruptions in the contracted supply chain. When a terminal is offline, buyers redirect spot cargoes from other terminals; sellers redirect cargoes from other markets. This mechanism functioned — imperfectly and at high cost — during the 2022 energy crisis.

The design basis attack is calibrated to exceed the spot market's adaptive capacity. Multiple simultaneous terminal disruptions, coordinated with tanker-route interdiction and European import disruption, remove the supply nodes and the transport pathway the spot market depends on for reallocation. An adversary who has studied the 2022 crisis knows exactly how much supply disruption the spot market can absorb — and designs the attack to exceed that threshold.

Spot market cascade effects include: price spikes that affect all LNG buyers globally, not only the directly targeted markets; cargo diversion from Asian buyers to European emergency procurement, creating supply stress in Asian markets that may affect allied supply security in the

Pacific theatre; and financial market stress in energy company securities and credit instruments, with secondary effects on infrastructure investment and project financing for the terminal replacement and expansion that recovery requires.

5.5 The Simultaneity Trap: Terminal Plus Pipeline

The most consequential version of the LNG terminal attack does not occur in isolation. It occurs as the terminal component of an integrated campaign that simultaneously executes the L2 Scenario, a natural gas pipeline cascade against the transmission infrastructure feeding the export terminals, and the L2 Scenario C pipeline rupture and ignition against product pipelines in the affected region.

In this integrated scenario, the export terminal is attacked from the maritime side while its gas supply feed is disrupted from the land side. Even if the terminal's liquefaction and storage systems are undamaged, the terminal cannot produce LNG without feed gas from the transmission system. The marine damage prevents loading operations. The cyber component undermines confidence in the safety system and extends the restart timeline. The subsea pipeline damage disrupts the alternative supply route that would otherwise allow the terminal to receive gas via an offshore pipeline interconnection.

Every adaptive mechanism the terminal operator might employ — switching feed gas sources, using onsite storage inventory, accepting emergency berthing by alternative carriers — has been anticipated and addressed by the coordinated attack design. This is the simultaneity trap applied to the LNG supply chain: each disruption is addressable; their simultaneous occurrence exceeds the system's adaptive capacity across every recovery pathway at once.

6. Protective Action Thresholds

6.1 Threat Detection and Indicator Taxonomy

6.1.1 Elevated Threat Indicators

- Intelligence Community reporting of adversary interest in LNG terminal infrastructure, shipping lanes, or terminal facility layouts.
- Anomalous ICS network activity: unusual DCS historian queries, lateral movement between corporate and OT network segments, anomalous vendor access sessions outside maintenance windows.
- Aerial UAS activity in terminal airspace or approach corridors outside established flight paths.
- Unidentified small surface vessels in terminal approach channels or security zones during hours inconsistent with commercial traffic patterns.

- Unusual subsea acoustic signatures in terminal approach channels or near subsea pipeline routes.
- Physical surveillance of terminal marine infrastructure from vessels or shoreside positions.
- Anomalous LNG tanker scheduling changes or cargo cancellations across multiple terminals within a short time window — a potential indicator of advance knowledge of supply disruption.

6.1.2 Imminent Threat Indicators

- Confirmed unauthorized aerial UAS in terminal airspace.
- Radar or visual contact with an unidentified small surface vessel on the approach vector to the terminal marine infrastructure.
- Simultaneous DCS anomalies at multiple terminals within a single operating company or across the Gulf Coast terminal cluster.
- Sonar contact with unidentified subsea object in terminal approach channel or near pipeline route.
- Confirmed physical intrusion at terminal facility perimeter.
- Loss of communication with SCADA systems at multiple terminals within a short time window.

6.2 Decision Authority Framework

- Facility level: Terminal operator authority to initiate emergency shutdown, vessel traffic suspension, and site security response. USCG notification is required for maritime security incidents.
- USCG: Maritime security zone establishment, vessel traffic control, and coordination with the Navy for maritime threat response. The Captain of the Port (COTP) authority for immediate maritime security actions.
- CISA / DOE: Federal cyber incident coordination and ESF-12 activation for supply disruption response.
- Department of Defense / Navy: Maritime threat response in territorial waters and beyond, in coordination with USCG. UUV and USV threat response may require Navy mine countermeasures or explosive ordnance disposal capabilities not available to civilian security agencies.
- National Security Council: Attribution assessment and response options for state-actor attacks. Alliance notification and coordination through NATO channels.

- DOE Export Authorization: Emergency procedures for supply redirection, including potential suspension or redirection of export authorizations to prioritize allied supply.

6.3 Bright Line Criteria

The following events constitute Bright Lines — thresholds triggering mandatory federal coordination and, where authority exists, federal response action:

- Confirmed kinetic attack on LNG terminal marine infrastructure at any U.S. export terminal.
- Confirmed DCS compromise with suspected nation-state attribution at any LNG terminal.
- Confirmed USV or UUV attack or attempted attack against terminal infrastructure or moored LNG carrier.
- Simultaneous security incidents at two or more U.S. LNG export terminals within a 24-hour window.
- Attack on allied LNG import terminal with confirmed or suspected state actor attribution — activating NATO energy security consultation procedures.
- LNG supply disruption affecting European allied supply commitments exceeding defined threshold volumes for a defined duration — triggering NATO energy security coordination mechanism.

7. Recovery Architecture

7.1 Terminal Recovery Sequencing

Phase 1 — Immediate (0-72 hours): Emergency shutdown and facility security. USCG maritime security zone. Federal incident coordination activation. ICS forensic investigation initiation. In-transit LNG cargo diversion to alternative terminals. Storage inventory drawdown at receiving terminals to maintain supply continuity: Allied government and NATO notification.

Phase 2 — Short-term (72 hours to 4 weeks): Damage assessment completion. Marine infrastructure repair scope defined. DCS forensic investigation complete and system reconstitution underway. Alternative U.S. terminal capacity maximized. Qatar, Norwegian, and Australian LNG supply redirection negotiated via emergency spot-market procurement. FSRU deployments in Europe accelerated, with additional units available.

Phase 3 — Medium-term (4 weeks to recovery): Loading arm procurement and installation. Liquefaction train restart following DCS reconstitution and safety system validation. Subsea pipeline repair. Incremental return to export operations as marine infrastructure is restored—European storage inventory rebuilding as supply resumes.

7.2 Alternative Supply Routing

Domestic terminal surge. Undamaged U.S. export terminals can be operated above normal throughput rates for limited periods to compensate for lost capacity partially. Practical limits are imposed by feed gas supply contracts, storage inventory, and vessel scheduling.

Qatari supply redirection. Qatar is the second-largest global LNG exporter and has significant flexible supply capacity. Emergency procurement from Qatari suppliers — at spot market prices elevated by the supply disruption — provides the most accessible alternative to U.S. supply for European buyers. U.S. government diplomatic coordination with Qatar is the relevant mechanism.

Norwegian pipeline maximization. Norwegian pipeline exports to the UK and continental Europe can be maximized within physical infrastructure limits. Norway has been operating near capacity since the 2022 crisis; additional capacity is limited but provides marginal relief.

Demand reduction. European member-state governments have developed demand-reduction frameworks during the 2022 crisis — industrial curtailment, heating-temperature regulations, and public conservation measures. These frameworks can be reactivated to reduce the supply shortfall that storage drawdown and alternative procurement must cover.

7.3 Allied Coordination Mechanisms

NATO Energy Security Consultation. NATO's energy security framework includes consultation mechanisms for supply disruptions that affect member states' security. These mechanisms were designed for peacetime supply crises; their activation in a military-action context — where the supply disruption is itself part of an adversary's military campaign — has not been fully exercised. The DBA-MA framework recommends the dedicated exercise of NATO energy security consultation procedures under conditions of military action.

EU Emergency Supply Measures. The EU Gas Security of Supply Regulation establishes demand-reduction and supply-solidarity mechanisms for emergency conditions. Activation under the military-action scenario requires coordination between EU emergency supply mechanisms and NATO defense planning processes — a coordination with no established operational protocol.

U.S.-EU Energy Security Dialogue. The U.S.-EU Energy Council and the transatlantic LNG supply relationship established since 2022 provide the bilateral diplomatic infrastructure for emergency supply coordination. Pre-agreed emergency protocols — defining trigger conditions, notification timelines, and supply redirection authorities — would reduce the decision latency that compounds the consequences of supply disruption in the first 72 hours of an event.

8. Gaps and Recommendations

8.1 Regulatory Gaps

USV and UUV threat gap in maritime security frameworks. MTSA-based facility security plans and USCG security zones were designed for conventional vessel threats. The unmanned surface and subsea vehicle threat — demonstrated by the Houthi maritime campaign and the Nord Stream attack — is not systematically addressed in current terminal security plan requirements. USCG should develop and promulgate specific USV/UUV threat assessment and countermeasure requirements for LNG facility security plans.

Aerial UAS threat gap. FAA airspace regulation and facility security perimeter controls are not integrated into a coherent counter-UAS framework for LNG terminal protection. Authority for counter-UAS action in terminal airspace is fragmented among the FAA, DHS, and DOD, creating response gaps. Legislative clarification of counter-UAS authority at critical energy infrastructure sites — following the model of the FAA Reauthorization Act's approach to airport counter-UAS — is needed.

DOE export authorization emergency gap. DOE's LNG export authorization framework lacks a pre-established emergency protocol for redirecting supply in a military action crisis. Pre-authorized emergency redirection authority, established by regulation or executive order before a crisis occurs, would reduce the response time for supply chain adaptation from weeks to days.

NATO-civilian framework integration gap. No established operational protocol integrates EU emergency supply mechanisms with NATO energy security consultation procedures under conditions of military action. This gap should be addressed through dedicated allied planning and exercise.

8.2 Physical and Maritime Hardening Priorities

Marine infrastructure hardening. Loading arm and jetty structures are the governing constraints on recovery for export terminal attacks. Investment in the physical protection of marine infrastructure — standoff barriers, hardened loading-arm mounting structures, and redundant berth capability — reduces both the consequences of a successful attack and the recovery timeline.

Subsea surveillance systems. Active sonar arrays, fiber-optic acoustic sensing systems, and periodic UUV inspection programs for terminal approach channels and subsea pipeline routes would provide detection capability against the UUV threat vector currently absent from most terminal security postures.

Loading arm spare parts strategic stockpiling. Loading arm replacement is the long-lead-time recovery constraint. Industry and government coordination on strategic stockpiling of critical loading arm components — analogous to the electricity sector's large power transformer spare equipment program — would reduce the marine infrastructure recovery timeline from 18 months to a shorter period for some damage configurations.

8.3 UAS Detection and Countermeasure Priorities

Integrated counter-UAS systems. Deployment of integrated aerial, surface, and subsea UAS detection systems at major LNG export terminals, combining radar, electro-optical, acoustic, and sonar sensing with centralized monitoring capability. Detection systems alone are insufficient without the legal authority and response capability to act on detections — the authority gap identified in Section 8.1 must be addressed concurrently.

Industry-USCG-Navy counter-UAS coordination. Terminal operators, USCG, and Navy explosive ordnance disposal and mine countermeasures units should develop and exercise joint response protocols for UUV and USV threats at LNG facilities. The Navy has capabilities relevant to the subsea UAS threat that USCG and terminal operators cannot replicate; pre-established coordination protocols reduce response time from hours to minutes.

License plate recognition and approach road surveillance. As documented in DBA-GAS-FC1, LPR systems on terminal approach roads — operated in partnership with local law enforcement — provide pre-attack reconnaissance detection capability. The nuclear sector's Turnaround Log precedent applies equally to LNG terminal land-side access: pattern recognition across multiple terminals, aggregated by the ONE-ISAC (formerly ONG-ISAC), can identify coordinated surveillance activity before an attack is executed.

8.4 Allied Coordination Recommendations

Dedicated NATO energy security exercise. NATO should conduct a dedicated exercise of energy security consultation and coordination procedures under military-action scenario conditions, specifically including LNG supply disruption as the energy stress event. The exercise should integrate member state military logistics planning with civilian emergency supply response to identify and address the coordination gaps documented in Section 7.3.

U.S.-EU pre-agreed emergency supply protocols. The transatlantic energy security relationship established since 2022 should be formalized into pre-agreed emergency protocols with defined trigger conditions, notification timelines, and supply redirection authorities. Protocols negotiated before a crisis are operationally superior to protocols negotiated during one.

FSRU security standards. The post-2022 FSRU deployments across European member states were executed rapidly, prioritizing supply capacity over security posture. A coordinated security standards program for FSRU facilities — developed jointly by EU energy security bodies, member state coast guards, and the maritime classification societies that certify FSRU vessels — would address the security gaps created by rapid deployment.

8.5 Research and Development Priorities

Rapid-deployment loading arm technology. Investment in modular, rapid-deployment loading arm designs that can be installed on standard jetty infrastructure within weeks rather than months would materially reduce the governing recovery constraint for export terminal attacks. This is the highest-priority R&D investment for LNG terminal recovery architecture.

Autonomous underwater vehicle detection systems. Current commercial harbor protection sonar systems were designed for diver and swimmer threats, not for the speed and acoustic signature characteristics of modern military-grade UUVs. Investment in UUV-optimized detection systems calibrated to the threat characteristics of current adversary systems is needed.

LNG terminal ICS anomaly detection. ONG-specific ICS anomaly detection capable of distinguishing adversary DCS manipulation from normal cryogenic process variation — particularly in the context of LNG liquefaction train control systems — would reduce the cyber-component detection window from the current weeks-to-months timeframe toward days. This is the cyber equivalent of the physical surveillance gap and warrants an investment priority equivalent.

9. Physical Consequence Envelope

9.1 Understanding the Hazard Mechanisms: BLEVE, VCE, and Pool Fire

Three distinct hazard mechanisms are relevant to LNG terminal attacks. They differ in sequence, spatial footprint, and dominant injury mechanism. Understanding the difference is prerequisite to designing protective measures, evacuation zones, and emergency response protocols that address the right threat for each scenario.

Boiling Liquid Expanding Vapor Explosion (BLEVE). The container fails first; the explosion follows. A BLEVE occurs when a vessel holding a cryogenic or pressurized liquid fails catastrophically. For LNG, this is the relevant scenario for pressurized satellite storage vessels, LNG transport tanker cargo tanks, or any containment system where a breach occurs under conditions that cause rapid, uncontrolled vaporization. The liquid flashes to vapor almost instantaneously — at a volume expansion ratio of approximately 600 to 1 for LNG. If the vapor cloud ignites, the result is a rising fireball accompanied by a blast wave radiating outward from the failure point. Structural fragments from the ruptured vessel are also a significant hazard, extending the lethal zone beyond the thermal and overpressure radii. A BLEVE is a point event: the explosion originates at the failed vessel and its effects radiate outward from that location.

Vapor Cloud Explosion (VCE). The cloud forms first; ignition follows. A VCE occurs when a large quantity of flammable vapor disperses into the surrounding atmosphere, mixes with air to form a flammable cloud, and is subsequently ignited — at the source, at a distant ignition point, or after the cloud has traveled some distance on the wind. The combustion front races through the cloud volume simultaneously. If the cloud encounters confinement — structures, equipment corridors, low terrain — or if there is sufficient turbulence, combustion can accelerate from a deflagration into a high-overpressure event with a destructive blast wave. Unlike a BLEVE, a VCE is a volume event: the explosion fills the space the cloud occupied, which may extend hundreds of meters from the spill point. The hazard zone is therefore determined not only by the volume of LNG released but by wind speed, wind direction, surface roughness, and the location and timing of ignition. A cloud that travels 300 meters before igniting produces a consequence zone centered 300 meters from the release point, not at the release point itself.

Pool Fire. LNG spills and ignites immediately at the release point, burning as a liquid pool. There is no explosion — the dominant hazard is intense thermal radiation over the pool area and downwind. Pool fires are the most common LNG ignition scenario and, for a given release volume, produce a more predictable and geometrically contained consequence envelope than VCE or BLEVE scenarios. They are not, however, benign: large-diameter pool fires at LNG terminal scale can produce thermal radiation intensities that cause fatalities at distances of several hundred meters, and the thermal flux at the pool perimeter makes firefighting approach without specialized equipment impossible.

9.2 Release Scenarios Under Military-Action Conditions

Four release scenarios are relevant to the DBA-MA LNG terminal attack. They are not mutually exclusive; the design-basis attack sequence described in Section 4 can initiate multiple scenarios simultaneously or in rapid succession.

9.2.1 Storage Tank Breach

Full-containment LNG storage tanks — the large cryogenic tanks that hold LNG between production and loading — are designed to resist external hazards including fire, seismic events, and aircraft impact. They are not designed to withstand shaped charge or kinetic penetrator attack. A successful breach of a storage tank's inner vessel releases LNG into the annular space between the inner and outer tank shells. Depending on the breach location, extent, and the outer shell's integrity, LNG may escape the secondary containment bund and spread across the terminal area.

A large-diameter LNG pool fire resulting from a storage tank breach is the most likely immediate consequence of a kinetic strike. Tank capacity ranging from 160,000 to 200,000 cubic meters represents a very large fuel inventory. If the secondary containment bund is intact, the pool fire is confined to the bund area. If the bund is breached — by the attack itself or by the fire's thermal effects on surrounding infrastructure — the LNG can spread beyond the contained area, extending the pool fire footprint and creating conditions for additional ignition of adjacent equipment and process areas.

The BLEVE scenario for a storage tank breach requires a specific sequence: breach of the tank's primary containment under conditions that trap LNG in a partially confined space that is subsequently heated, causing pressure buildup followed by catastrophic failure. Full-containment tanks are specifically designed to prevent this sequence; the risk is higher at smaller satellite storage tanks, pressurized LNG transport vehicles, or in the marine context, aboard LNG carrier cargo tanks during a midships strike while the vessel is alongside at the berth.

9.2.2 LNG Carrier Strike

An LNG carrier alongside at the berth, with cargo tanks holding 140,000 to 266,000 cubic meters of LNG, represents a very high-consequence target for a kinetic strike. The cargo tanks aboard modern LNG carriers are membrane or Moss-type spherical tanks. Both designs have significant resistance to external impact in normal maritime operations; neither is designed for military-grade shaped charge or explosive attack.

A USV attack against a carrier's hull at the waterline — the Houthi operational template applied to an LNG carrier — would breach the hull and potentially the cargo tank, releasing LNG into the sea. LNG on water vaporizes rapidly, creating a vapor cloud that may extend over a large area of the water surface and the adjacent jetty. If the vapor cloud reaches an ignition source before dispersing — the terminal's own process equipment, the jetty electrical systems, or the engine room of the carrier itself — the result is a VCE or flash fire over the water surface and the terminal marine area simultaneously.

A strike against the cargo area of a fully laden carrier alongside the berth is the highest-consequence physical scenario in the LNG terminal attack design basis. The combination of a large LNG inventory, proximity to terminal infrastructure, restricted water area limiting vapor dispersion, and the simultaneous degradation of terminal safety systems through the cyber component of the attack creates conditions for a consequence envelope that exceeds the design basis of the terminal's emergency response plan.

9.2.3 Liquefaction Train Process Excursion

The DCS manipulation component of the design-basis attack can drive liquefaction train process parameters outside safe operating limits. The cryogenic process envelope is narrow: the refrigerant circuits, heat exchangers, and compressor trains operate within tightly bounded temperature and pressure ranges. Manipulation of process setpoints — heat exchanger bypass valves, refrigerant composition, compressor suction pressures — can cause brittle fracture of cryogenic piping, compressor surge, or overpressure conditions in process vessels. These failure modes can release process fluids, natural gas, or refrigerant gases into the process area.

A process gas release in the liquefaction area, combined with ESD inhibit commands from the cyber component and suppression of fire and gas detection alarms, creates conditions for vapor cloud formation in the process equipment area. Ignition sources are abundant in an operating liquefaction facility. A VCE in the process area — among close-spaced process equipment, pipe racks, and vessels — would produce blast overpressures significantly higher than the same cloud volume in open terrain, because the confinement and turbulence from the equipment geometry accelerates the combustion front.

9.3 Consequence Envelopes

Physical consequence envelopes for LNG releases are calculated using validated dispersion and consequence models; specific hazard distances for individual terminals require site-specific modeling with actual tank sizes, terrain, and weather data. The ranges below represent order-of-magnitude planning parameters for the scenario types defined in this section, derived from publicly available regulatory guidance including FERC siting requirements, PHMSA consequence analysis protocols, and USCG thermal exclusion zone calculations.

Thermal radiation — pool fire. For a large-diameter pool fire confined to a secondary containment bund at a major storage tank, the thermal radiation hazard zone extends from approximately 200 to 500 meters from the pool edge, depending on pool diameter and atmospheric conditions. The 5 kW/m^2 thermal flux contour — the threshold for pain and the onset of injury for unprotected individuals — typically falls at 300 to 600 meters from the pool center for bund-contained fires at large terminal tanks. The 37.5 kW/m^2 contour — the threshold for structural damage and immediate fatality for unprotected individuals — falls within the bund area itself for most large-tank fire scenarios. Personnel inside the terminal fence during a large pool fire are at high risk; personnel in the surrounding community are at risk within a radius that, for the largest terminal tanks, extends to several hundred meters beyond the fence line.

Flammable vapor cloud — VCE precursor. Before a VCE ignites, the flammable vapor cloud defines the zone within which an explosion can occur. For LNG, the lower flammable limit (LFL) is approximately 5% methane in air; the upper flammable limit (UFL) is approximately 15%. The cloud's flammable zone — the area between the LFL and UFL concentration contours — is the consequence-relevant region. For a large spill from a storage tank, the flammable vapor cloud can extend hundreds of meters downwind under stable atmospheric conditions before dispersing below the LFL. FERC siting regulations require LNG facilities to demonstrate that the flammable vapor cloud from a maximum credible spill does not reach offsite ignition sources; the design-basis military-action spill may exceed the maximum credible spill volume used in the original siting analysis.

Blast overpressure — VCE. The blast overpressure from a VCE depends on the volume of the flammable cloud, the degree of confinement, and the rate of combustion. In open terrain, a VCE from a large LNG spill may produce overpressures of 0.1 to 0.3 bar (1.5 to 4.5 psi) at distances of 100 to 300 meters from the cloud center, causing window glass failure and structural damage to unreinforced buildings. In a congested process equipment area, the same cloud volume can produce significantly higher overpressures due to the acceleration of the combustion front by equipment geometry. The process area of a liquefaction train is among the most congested environments at an LNG terminal, which is why a DCS-manipulation-induced VCE in the process area is a higher-overpressure event than the same release in an open storage area.

BLEVE fireball and blast. A BLEVE from a pressurized LNG storage vessel or a carrier cargo tank produces a fireball whose radius scales with the cube root of the LNG mass involved, and a blast wave that attenuates with distance from the burst point. For a large LNG carrier cargo tank (approximately 25,000 to 40,000 cubic meters per tank), the fireball radius can reach several hundred meters and persist for tens of seconds. The thermal dose delivered by a BLEVE fireball — the product of radiation intensity and duration — is the primary fatality mechanism at distances beyond the blast overpressure zone. Fragment hazard from the ruptured vessel extends the lethal zone further, with heavy fragments potentially traveling hundreds of meters.

9.4 Physical Consequence Under Military-Action Conditions

The physical consequence envelopes described in Section 9.3 assume that emergency response can be initiated promptly. Standard LNG facility emergency response plans establish evacuation zones, firefighting resource positioning, and USCG maritime security zone activation based on an assumed response timeline of minutes to tens of minutes. Military-action conditions violate this assumption in three ways that materially expand the physical consequence envelope.

Degraded detection. The cyber component of the design-basis attack suppresses fire and gas detection alarms and inhibits ESD actuation. Emergency response cannot begin until operators independently confirm the initiating event through means other than the compromised alarm system. The delay between the initiating event and confirmed detection extends the duration of uncontrolled LNG release before emergency response actions are taken, increasing the pool size, vapor cloud extent, and the thermal dose delivered to terminal personnel who have not yet evacuated.

Denied access. The local sovereignty principle established in the DBA-MA parent framework — that protective systems must be capable of operating without external access — applies with particular force to LNG terminal physical consequence management. Firefighting at a large LNG pool fire requires approach to within the thermal hazard zone; under active military threat conditions, fire service personnel cannot safely approach. The fire burns to depletion of the available LNG inventory rather than being suppressed. For a large storage tank fire, this extends the duration of thermal radiation exposure for the surrounding community from the period of firefighting response to the hours-long period required for the tank inventory to exhaust.

Compounded simultaneous events. The design-basis attack initiates multiple release scenarios simultaneously: the maritime USV strike against the jetty structure, the DCS manipulation of the liquefaction train, and the aerial UAS strike against control infrastructure all occur within a 15-minute window. Emergency response resources — terminal fire brigades, USCG, local mutual aid — are not sized to address simultaneous events at multiple locations within the same facility. The consequence envelope of the combined event exceeds the sum of the individual events, because the resources that would address one scenario are unavailable to address the others.

The practical implication for emergency planning is that LNG terminal emergency response plans must establish a “military-action scenario” tier alongside the existing all-hazards planning tier. This tier acknowledges that the nominal response timeline cannot be achieved, that access to the terminal may be denied for some period, and that simultaneous events at multiple terminal locations must be managed with reduced response capability. Community consequence planning

— evacuation zones, shelter-in-place protocols, hospital system surge planning — must be sized for the military-action scenario consequence envelope, not the standard emergency response scenario.

9.5 Implications for Facility Security Planning and Regulatory Requirements

Current LNG terminal facility security plans, prepared under USCG and FERC requirements, address the physical consequence of LNG releases in terms consistent with the existing regulatory siting and safety requirements. Those requirements were developed for the all-hazards planning baseline: accidental releases, equipment failures, and natural hazards. The design-basis military-action event described in this section exceeds the assumed release volume, the assumed response timeline, and the assumed response access conditions that underlie the existing regulatory consequence analysis.

Three specific regulatory gaps follow from this analysis. First, FERC siting consequence analysis for LNG terminals should be extended to address military-action scenario release volumes, not only the maximum credible accidental spill. Second, USCG facility security plan requirements should mandate a military-action scenario consequence analysis tier, including community consequence planning for the extended consequence envelope under denied-access conditions. Third, the physical consequence analysis in this section should inform the counter-UAS and maritime hardening priorities identified in Section 8.3 — the return on investment for pre-attack detection and hardening measures is materially higher when the avoided consequence is a military-action scenario event rather than a standard accidental release.

Appendix A: LNG Terminal Strategic Inventory

The following table provides an expanded reference inventory of LNG terminals relevant to the U.S./NATO strategic supply chain analyzed in this document.

Terminal	Country	Type	Capacity (mtpa)	NATO/Allied Significance
Sabine Pass	USA	Export	~30	Largest U.S. export terminal; primary European supply under long-term contracts
Freeport LNG	USA	Export	~15	Second largest; the 2022 outage demonstrated European market sensitivity
Corpus Christi	USA	Export	~15	Significant European contract base; expansion capacity
Cameron LNG	USA	Export	~12	European and Asian contracts, including French and Japanese buyers

Terminal	Country	Type	Capacity (mtpa)	NATO/Allied Significance
Cove Point	USA	Export	~5.5	Sole U.S. Atlantic Coast export terminal; direct European pathway
Elba Island	USA	Export	~2.5	Smaller capacity; Southeast U.S. coastal location
LNG Canada (Ph1)	Canada	Export	~14	NATO ally; Pacific Coast; Asia-Pacific primary market
Gate Terminal	Netherlands	Import	~16	Primary NW European hub; Germany, Belgium, Netherlands gateway
Zeebrugge	Belgium	Import/Trans	~9	Major trans-shipment hub; UK and continental Europe
Świnoujście	Poland	Import	~6.2	Critical NATO eastern flank; expansion underway
Eemshaven FSRU (DE)	Germany	FSRU Import	~5	Post-2022 emergency deployment; significant vulnerability profile
Brunsbüttel FSRU (DE)	Germany	FSRU Import	~5	Post-2022 emergency deployment
Piombino FSRU (IT)	Italy	FSRU Import	~5	Post-2022 emergency deployment; Mediterranean supply
Inkoo FSRU (FI)	Finland	FSRU Import	~5	NATO northern flank; Baltic region supply security

Appendix B: Regulatory Authority Crosswalk

Agency/Body	Jurisdiction	Security Authority	Gap Relevance
USCG	Maritime security, 33 CFR 127, MTSA facility security plans	Strong physical/maritime; facility security plan review and enforcement	USV/UUV threat not addressed in current security plan standards
FERC	LNG terminal siting, construction, operation (Natural Gas Act)	Safety review of system design; no direct security mandate	No security oversight of operational facilities post-construction
TSA	Surface transportation security	Landside coordination is less direct than the pipeline context	Counter-UAS authority gap for landside approaches
PHMSA	Onshore LNG storage, 49 CFR 193	Safety-focused; no military-action threat tier	Safety/security integration gap
DOE	Export authorization; ESF-12; SPR (crude/products only)	Emergency response coordination; export redirection authority	No pre-authorized emergency export redirection protocol

Agency/Body	Jurisdiction	Security Authority	Gap Relevance
FAA	Airspace management	No security mandate; UAS traffic management	Counter-UAS authority gap at the terminal airspace
DOD/Navy	Territorial waters defense; EOD/mine countermeasures	Military response authority; UUV/USV response capability	No routine coordination protocol with terminal operators
NATO	Alliance defense planning; energy security consultation	No direct facility jurisdiction; alliance coordination framework	No military action scenario, energy security exercise protocol
EU Commission	Gas Security of Supply Regulation; member state solidarity	Supply standard obligations; demand reduction frameworks	Military-action scenario exceeds regulatory design basis

Appendix C: Acronyms

BLEVE	Boiling Liquid Expanding Vapor Explosion
VCE	Vapor Cloud Explosion
CISA	Cybersecurity and Infrastructure Security Agency
COTP	Captain of the Port (USCG)
DBA-MA	Design Basis Accident — Military Action
DCS	Distributed Control System
DNG-ISAC	Downstream Natural Gas Information Sharing and Analysis Center
DOD	U.S. Department of Defense
DOE	U.S. Department of Energy
EOD	Explosive Ordnance Disposal
ESD	Emergency Shutdown System
ESF-12	Emergency Support Function 12 (Energy)
EU	European Union
FAA	Federal Aviation Administration
FERC	Federal Energy Regulatory Commission
FSRU	Floating Storage and Regasification Unit
ICS	Industrial Control System
LNG	Liquefied Natural Gas
LPR	License Plate Recognition
LLEA	Local Law Enforcement Agency
MTSA	Maritime Transportation Security Act

NATO	North Atlantic Treaty Organization
ONE-ISAC	Oil and Natural Gas Information Sharing and Analysis Center
ONG SCC	Oil and Natural Gas Subsector Coordinating Council
PHMSA	Pipeline and Hazardous Materials Safety Administration
SCADA	Supervisory Control and Data Acquisition
SDV	Swimmer Delivery Vehicle
SPR	Strategic Petroleum Reserve
TSA	Transportation Security Administration
UAS	Unmanned Aerial/Aquatic System (collective)
USCG	United States Coast Guard
USV	Uncrewed Surface Vehicle
UUV	Uncrewed Underwater Vehicle

Revision History

Version	Date	Author	Description
1.0	April 2026	Tim Roxey	Initial release. Strategic supply chain consequence analysis for deliberate military action against LNG terminal infrastructure. Physical consequence envelope (BLEVE, VCE, pool fire) deferred to companion document. Issued under DBA-MA Oil and Natural Gas Child Document.
1.1	April 2026	Tim Roxey	Repositioned as DBA-GAS-UC-LNG under the DBA-GAS Sector Framework. Section 9 (Physical Consequence Envelope) added: BLEVE, VCE, and pool fire mechanisms explained; storage tank breach, LNG carrier strike, and liquefaction train process excursion scenarios developed; consequence envelopes defined; military-action condition amplifiers addressed; regulatory gap implications identified. VCE added to acronym table. ONG-ISAC updated to ONE-ISAC. Parent references updated throughout. Revision history added.

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